

AI Gene Review — Focused Hypothesis Report

K9IMD0 (TRLF_DESRO), *Desmodus rotundus* lactotransferrin / draculin

Focus: function_assignment — "lactotransferrin functions and draculin specialization"

Executive Judgment

Verdict: PARTIALLY SUPPORTED / largely OVER-ANNOTATED (family carry-over).

K9IMD0 is unambiguously **draculin**, and draculin has been explicitly established as "*a mutated version of the lactotransferrin scaffold*" (P 23748026).

The seed hypothesis asks whether this protein still *directly* holds a battery of inherited transferrin/lactoferrin GO terms independent of its anticoagulant specialization. The evidence splits sharply:

- **Experimentally supported for THIS protein (retain, core):** secretion / extracellular region (GO:0005576, IDA) and **negative regulation of blood coagulation** (GO:0030195, IDA) — FXa/FIXa inhibition

(P 9795244),

(P 10556567).

Anticoagulation is the *primary, demonstrated* function.

- **Structurally defensible but not assayed in bat (retain only as ISS/IEA with a species caveat):** metal-ion binding (GO:0046872) — the **N-lobe iron site is fully conserved**; and

serine-type peptidase (GO:0008236) — the **nucleophile Ser is conserved** but one essential catalytic residue is altered (Lys→Arg).

- **Weak / electronic carry-over (generalize, flag non-core, or remove pending evidence):** iron ion transport (GO:0006826/0006811) — the **C-lobe iron site is broken (Y454→D452)** so bilobal transport is degraded; antibacterial humoral response / defense-to-bacterium / immune-system-process (GO:0019731/0042742/0002376); and the endosome / recycling-endosome / plasma-membrane locations (GO:0005769/0055037/0005886), which derive from mammalian lactoferrin **receptor recycling** and have no support for a secreted salivary bat protein.

Most important caveat: every non-anticoagulant term is IEA:TreeGraft or a family keyword — none rests on a direct assay of draculin. The seed's framing is correct that "anticoagulant specialization does not exclude inherited functions," but the molecular evidence shows *active degradation* of two ancestral active sites (C-lobe iron Tyr; catalytic Lys of the protease dyad), so inheritance cannot be assumed and should be curated conservatively.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
P 23748026	Structural/evolutionary (proteome+transcriptome)	Qualifies (identity)	K9IMD0 = purified draculin	Draculin "established... a mutated version of the lactotransferrin scaffold"; large sequence stretch recovered
P 7740503	Direct assay (purification)	Supports (secreted anticoagulant)	Draculin is a secreted salivary anticoagulant glycoprotein	Purified ~88 kDa glycoprotein the saliva anticoagulant factor
P 9795244	Direct assay (biochemistry)	Supports (core function)	Anticoagulation	Inhibits FIXa and FXa; glycosylation-dependent activation
P 10556567	Direct assay (kinetics)	Supports (core function)	FXa inhibition mechanism	Tight-binding, noncompetitive FXa inhibitor, $K_i \approx 14.8$ nM
P 23411029	Localization/expression (RNA-seq + LC-MS/MS)	Qualifies	Where/what class	"Lactotransferrin" among accessory-gland antimicrobials ; secreted
This report (align vs P02788)	Computational (structural/evolutionary)	Qualifies/ partly refutes	Iron-binding retained?	N-lobe Fe ligands (D79,Y111,Y211,H272)+carbon conserved; C-lobe Y454→D455 lost ; 72% id
P 12535064	Direct assay + mutagenesis	Qualifies	LF serine- protease mechanism	Catalytic dyad Ser259+Lys73 ; mutating either "dramatic decrease in proteolysis"
P 9770539	Direct assay	Qualifies	LF antibacterial protease	N-lobe serine-protease activity cleaves <i>H. influenzae</i> IgA1 protease/Hap; blocked by serine protease inhibitors
P 11163480	Direct assay	Qualifies	LF proteolysis site	LF cleaves Iga within the β - anchor region

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding
This report (dyad mapping)	Computational	Qualifies/ partly refutes	Protease dyad conserved in draculin?	Ser278 conserved; Lys92→Arg (essential Lys altered); cationic N-term net +9

GO Curation Implications (leads — require curator verification)

See [K9IMD0_GO_decision_table.csv](#) (computed provenance). Summary:

- **RETAIN (core, experimental):** GO:0005576 extracellular region (CC, IDA); GO:0030195 negative regulation of blood coagulation (BP, IDA). These are the protein's demonstrated identity and primary function.
- **RETAIN as ISS/IEA with explicit "not verified in *D. rotundus*" caveat, treat as non-core:** GO:0046872 metal ion binding (N-lobe site intact — a *bona fide* structural feature); GO:0008236 serine-type peptidase (nucleophile Ser conserved).
- **GENERALIZE / flag non-core / candidate for removal pending evidence:** GO:0006826 & GO:0006811 iron/ion transport (C-lobe site broken; no bat transport assay; a secreted protein without the receptor-recycling context); GO:0008233 peptidase, GO:0016787 hydrolase, GO:0006508 proteolysis (downstream/too-broad parents of an unverified activity); GO:0019731/0042742/0002376 antibacterial/defense/immune (family carry-over + transcriptomic grouping only).
- **REMOVE or down-weight to peripheral-IEA:** GO:0005769 early endosome, GO:0055037 recycling endosome, GO:0005886 plasma membrane — these describe the mammalian lactoferrin-receptor recycling itinerary and are not supported for a secreted salivary bat protein. (The seed's point that "missing TM topology does not exclude peripheral association" is fair, but there is *no positive evidence* of a membrane/endosomal pool of draculin.)

We avoid recommending "protein binding" as a terminal term; the informative core terms are the anticoagulation BP and extracellular CC.

Mechanistic Scope

- **Immediate molecular function (demonstrated):** noncompetitive tight-binding inhibition of activated coagulation factors Xa and IXa ($K_i \approx 15$ nM), requiring correct glycosylation. This is the direct gene-product activity.
- **Inherited scaffold features (molecular, partly degraded):** a bilobal transferrin fold with an intact N-lobe metal-binding pocket and a broken C-lobe pocket; a conserved protease nucleophile Ser but altered catalytic Lys.
- **Downstream / inferred-only (not direct activity):** iron *transport* (a systemic/cellular process needing receptor recycling), antibacterial *response*, and endosomal/plasma-membrane *localization* — all inferred from human lactoferrin biology, not observed for draculin.

Conflicts and Alternatives

- **Database carry-over (main risk):** TreeGrafter/PANTHER propagation and keyword→GO mapping assign the full lactoferrin repertoire to a protein the same authors call "mutated." Direct measurements exist only for anticoagulation.
- **Organism/paralog specificity:** all protease/antibacterial/iron-transport measurements are on **human** lactoferrin
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P 12535064

 / 9770539/11163480), not draculin. Two catalytic-site substitutions (C-lobe iron Tyr; protease Lys) argue against naive transfer.
- **Interpretation of the cationic N-terminus:** draculin retains a lactoferricin-like basic N-terminal stretch (net +9), so a peptide-based antimicrobial activity is *plausible* — a genuine alternative route to GO:0019731 that does not depend on the protease dyad. This remains untested.

Knowledge Gaps

1. **Does draculin bind iron?** Checked: sequence/structure only (N-lobe intact, C-lobe Tyr lost). Matters because it gates GO:0046872/0006826. Resolve with a spectrophotometric Fe^{3+} -saturation / urea-gel iron-binding assay on recombinant or salivary draculin.
2. **Does draculin have protease activity?** Checked: dyad mapping (Ser conserved, Lys→Arg). Matters for GO:0008236/0008233/0016787/0006508 and mechanistically for antibacterial. Resolve with an *H. influenzae* IgA1-protease/Hap cleavage assay ± serine-protease inhibitors, as in

P 9770539

3. **Is draculin antibacterial?** Checked: only transcriptomic grouping

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P 23411029

Resolve with MIC/killing assays vs Gram+/- bacteria.

4. **Any endosomal/plasma-membrane pool?** Checked: none found; IEA only. Resolve with cell-surface receptor-binding / uptake assays in a relevant cell type.
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Discriminating Tests

- **Iron binding:** UV-vis (465 nm holo-transferrin band) and urea-PAGE mobility shift on apo vs Fe-loaded recombinant draculin; compare N-lobe-only vs full-length.
 - **Protease:** recombinant draculin (and a draculin-R92K back-mutant) against IgA1-protease/Hap substrates; serine-protease-inhibitor panel.
 - **Antibacterial:** killing/MIC assays and a draculin N-terminal peptide (lactoferricin-analog) test to separate peptide- vs protease-based activity.
 - **Localization:** anti-draculin surface staining / receptor pulldown to test any membrane/endosomal association.
 - **Comparative genomics:** map the C-lobe Y→D and protease K→R substitutions across bat lactoferrins to test whether they are draculin-lineage specialization vs species polymorphism.
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Curation Leads (require curator verification)

Candidate references to attach (with exact snippets to verify): -

P 23748026

— verify: "...a very large sequence stretch of draculin and thus established that it is a mutated version of the lactotransferrin scaffold" (identity of K9IMD0 = draculin; supports functional divergence). -

P 12535064

— verify: "mutation of either Ser259 or Lys73 results in a dramatic decrease in proteolysis" (basis of the serine-protease term; note Lys→Arg in bat). -

P 9770539

— verify: "...localized to the N-lobe of the bilobed lactoferrin molecule and were inhibited by serine protease inhibitors..." (protease↔antibacterial link). -

P 9795244

P 10556567

— anticoagulant core function (FIXa/FXa inhibition). -

P 23411029

— lactotransferrin among accessory-gland antimicrobials (localization/expression, review-of-glands level).

Candidate action changes: - Keep GO:0030195 and GO:0005576 as **core/experimental**. - Downgrade GO:0006826/0006811, GO:0008233/0016787/0006508, GO:0019731/0042742/0002376 to **non-core, IEA/ISS with caveats**; consider removing the CC endosome/PM terms (GO:0005769/0055037/0005886) unless localization evidence is produced. - Where retained (GO:0046872 metal binding; GO:0008236 serine peptidase), annotate with an explicit note that the site is **structurally present but one essential residue diverges** and the activity is **unverified in *D. rotundus***.

Suggested curator questions: Is species-blinded IEA propagation appropriate for a protein the source literature calls "mutated"? Should the C-lobe iron-site loss and protease Lys→Arg be captured as sequence-feature caveats on the MF terms?

Limitations

All non-anticoagulant conclusions rest on sequence/structure inference and human-lactoferrin literature; no functional assay of draculin's iron binding, protease, or antibacterial activity exists. Alignment-based ligand mapping (72% identity to P02788) is robust for conserved columns but a single-substitution functional prediction (e.g., Lys→Arg) is a hypothesis, not a measurement.

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